

ECON 108 Discussion Section slope

Week 3. Multiple Linear Regression Model & Nonlinear Models

$$y = \beta_0 + \beta_1 x + u \rightarrow \frac{\Delta y}{\Delta x} = \beta_1$$

- **Agenda for week 3:** Expand a simple linear regression model by adding additional variables and incorporating nonlinear functions for the dependent and/or explanatory variables.

- **Shapes of a Quadratic Model:** $y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$ $\frac{\Delta y}{\Delta x} = \frac{dy}{dx} = \beta_1 + 2\beta_2 x$

The turnaround point?

$$\frac{\Delta y}{\Delta x} = 0 \Rightarrow \beta_1 + 2\beta_2 x = 0$$

How to tell the shape of the model based on the signs of coefficients? (Or vice versa.)

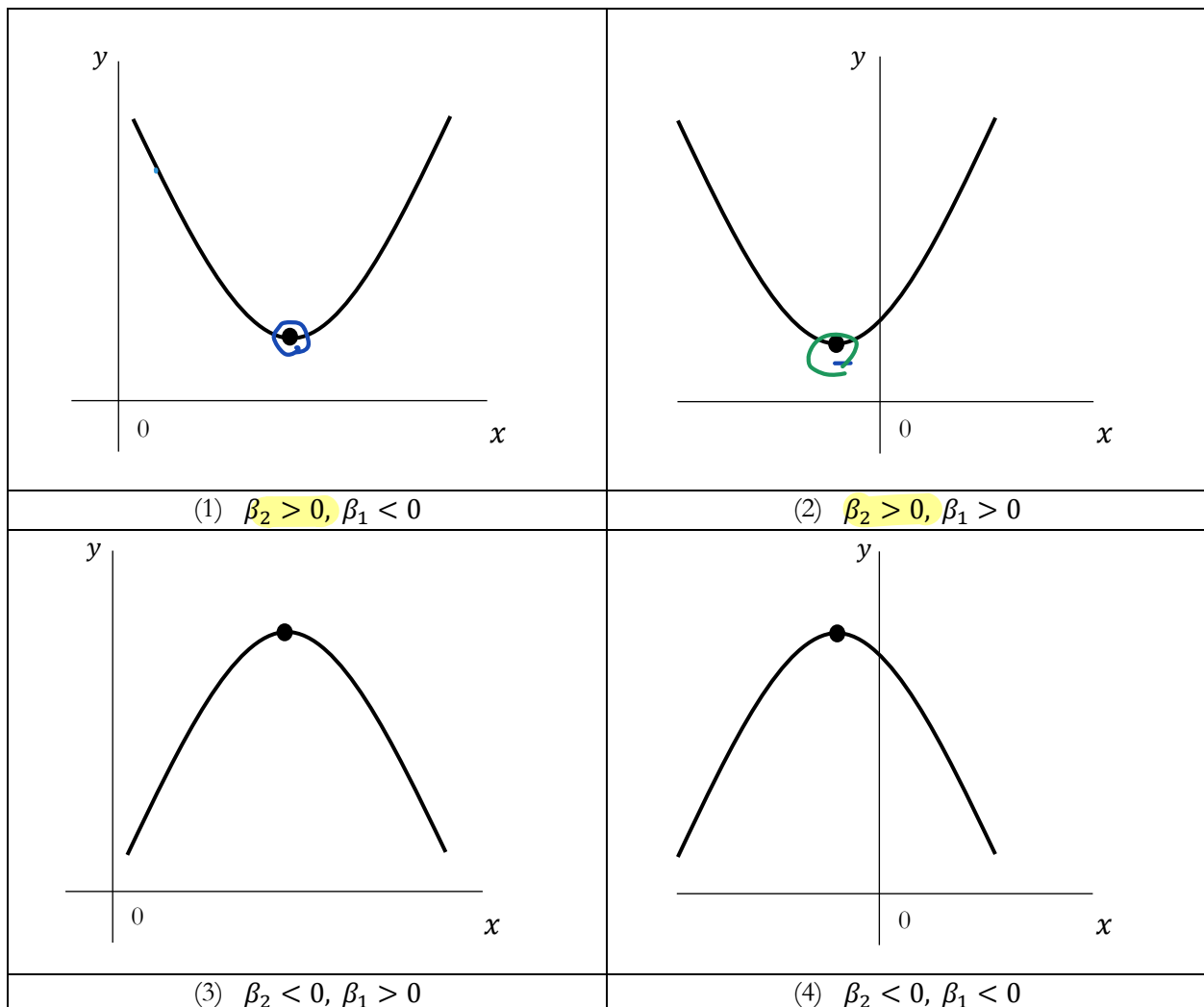
1. Check the sign of coefficient for x^2 (β_2)

- a. $\beta_2 > 0$: a U-shaped curve (1) and (2)
- b. $\beta_2 < 0$: an inverted U-shape curve (3) and (4)

2. Check the coefficient for x (β_1):

- a. If β_1 and β_2 have the same sign, the turning point is negative. (2) and (4)
- b. If β_1 and β_2 have the opposite signs, the turning point is positive. (1) and (3)

$$x^* = -\frac{\beta_1}{2\beta_2}$$



- Testing the nonlinear relationship between y and x in a quadratic model:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$$

$$y = \beta_0 + \beta_1 x + u$$

$H_0: \beta_2 = 0$ (i.e., no need for a quadratic term)

$H_a: \beta_2 \neq 0$ (i.e., the quadratic term is necessary)

- Interpreting nonlinear regression models

Case	Regression	Interpretation	
lin-lin	$y_i = \beta_0 + \beta_1 x_i + u_i$ $\Delta y = \beta_1 \Delta x$	A change in x by one unit is associated with a change in y by β_1 (unit), on average.	x : unit of x y : unit of y
lin-log	$y_i = \beta_0 + \beta_1 \ln x_i + u_i$	A 1% change in x is associated with $0.01 \cdot \beta_1$ change in y , on average.	x : % y : unit
log-lin	$\ln y_i = \beta_0 + \beta_1 x_i + u_i$	A change in x by one unit is associated with a $100 \cdot \beta_1$ % change in y , on average.	x : unit of x y : %
log-log	$\ln y_i = \beta_0 + \beta_1 \ln x_i + u_i$	A 1% change in x is associated with a β_1 % change in y , on average. β_1 is the elasticity of y with respect to X .	x : % y : %

lin-log

$$\Delta y = \beta_1 \cdot \Delta \ln x \times 100$$

$$\Delta y \cdot 100 = \beta_1 \cdot \% \Delta x$$

$$\Delta y = \left(\frac{\beta_1}{100} \right) \% \Delta x$$

log-lin

$$\Delta \ln y = \beta_1 \Delta x \times 100$$

$$\% \Delta y = (\beta_1 \times 100) \Delta x$$

Useful fact

$$\Delta \ln x \times 100 = \% \Delta x$$

log-log

$$\Delta \ln y = \beta_1 \cdot \Delta \ln x \times 100$$

$$\% \Delta y = \beta_1 \cdot \% \Delta x$$

Question 1.

Consider the following regression model that examines the relationship between the number of bedrooms and housing price:

$$Price_i = \beta_0 + \beta_1 Room_i + \beta_2 Room_i^2 + u_i$$

- a. What is the marginal effect of an additional bedroom on housing price?

$$\frac{\Delta Price}{\Delta Room} = \frac{d Price}{d Room} = \beta_1 + 2\beta_2 \cdot Room$$

- b. What is the predicted marginal effect of an additional bedroom on housing price for a house with 2 bedrooms = $Room$

$$\begin{aligned} \frac{\Delta Price}{\Delta Room} &= \beta_1 + 2\beta_2 \cdot \boxed{Room}^{=2} \\ &= \beta_1 + 4\beta_2 \end{aligned}$$

- c. Suppose we run an alternative model as follows. Interpret the coefficient on $\ln(Size_i)$. The unit of Size is square feet.

log-log $\ln(Price_i) = \alpha_0 + \alpha_1 \ln(Size_i) + \alpha_2 Room_i + u_i$

if the size of a house increases by 1%,
the housing price changes by α_1 %.

Question 2.

For the following scenario, predict the signs of the coefficients. Trace out the shape of the relationship between the dependent variable and the explanatory variable.

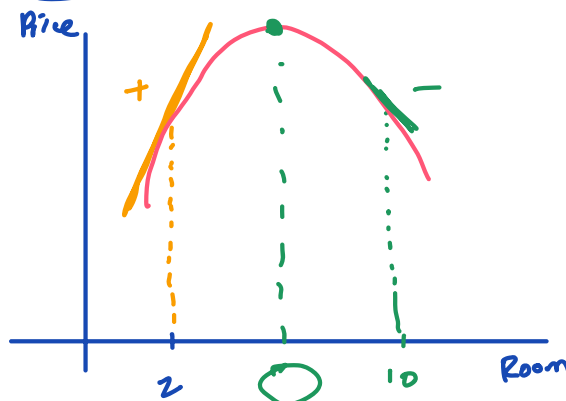
$$Price = \beta_0 + \beta_1 Room + \beta_2 Room^2 + u$$

Slope → The change in housing price from adding one room is \$50,000 when the current house has 2 bedrooms. The change in housing price from adding one room is -\$10,000 when the current house has 10 bedrooms.

$$\frac{\Delta Price}{\Delta Room} = \frac{d Price}{d Room} = \beta_1 + 2\beta_2 Room$$

slope

$$\text{slope} \begin{cases} 50,000 & \text{at } Room = 2 \\ -10,000 & \text{at } Room = 10 \end{cases}$$



① $\beta_2 < 0$ b/c the slope changes from (+) to (-)

② $\beta_1 > 0$ b/c the turning pt is a ³positive number (somewhere btw 2 & 10), and $\beta_2 < 0$. For the turning pt to be positive, β_1 & β_2 must have opposite signs.

Question 3.

Our research on examining the relationship between the number of computers per student and average test scores continues this week. This time, we include the computer-student ratio (comp_stu) and its squared term (comp_stu2) to investigate whether the relationship between the computer-student ratio and test scores (testscr) is nonlinear. Additionally, the model controls for the student-teacher ratio (str). Use the following Stata results to answer the questions.

```
. gen comp_stu2 = comp_stu^2
```

```
. reg testscr comp_stu comp_stu2 str
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$$\text{Score}_i = \beta_0 + \beta_1 \text{CSR}_i + \beta_2 \text{CSR}_i^2 + \beta_3 \text{STR}_i + u_i$$

Source	SS	df	MS	Number of obs	=	420
Model	15164.5307	3	5054.84358	F(3, 416)	=	15.36
Residual	136945.063	416	329.194863	Prob > F	=	0.0000
				R-squared	=	0.0997
				Adj R-squared	=	0.0932
Total	152109.594	419	363.030056	Root MSE	=	18.144

testscr	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
comp_stu	123.1777	46.28552	2.66	0.008	32.19501	214.1603
comp_stu2	-173.3259	131.4739	-1.32	0.188	-431.7619	85.11018
str	-1.533944	.4943405	-3.10	0.002	-2.505661	-.5622273
_cons	671.4727	11.16598	60.14	0.000	649.5239	693.4214

- a. Interpret the coefficient of the student-teacher ratio (str).

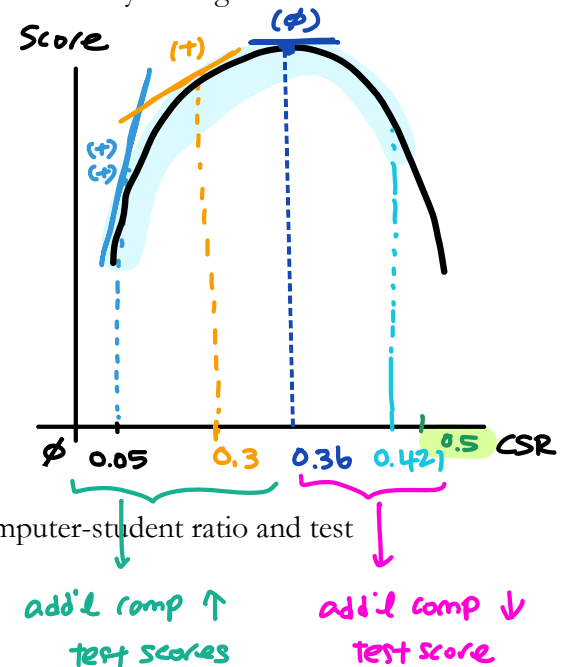
$\hat{\beta}_3 = -1.53$ "An add'l student per class decreases test scores by 1.53 points, all else equal."

- b. The computer-student ratio in the Anaheim Elementary School District is 0.05. What is the <predicted change in test scores> if the district purchases one additional computer per student?

$$\begin{aligned} \frac{\Delta \text{Score}}{\Delta \text{CSR}} &= \frac{\partial \text{Score}}{\partial \text{CSR}} = \hat{\beta}_1 + 2\hat{\beta}_2 \text{CSR}^{0.05} \\ &= 123.18 + 2 \cdot (-173.33) (0.05) \\ &= \underline{105.85} \end{aligned}$$

- c. The computer-student ratio in the Portola Valley Elementary School District is 0.3. What is the predicted change in test scores if the district purchases one additional computer per student? Will the change in test scores for Portola Valley Elementary be larger or smaller compared to your answer in part b?

$$\begin{aligned}\frac{\Delta \text{Score}}{\Delta \text{CSR}} &= \hat{\beta}_1 + 2\hat{\beta}_2 \cdot \text{CSR} \\ &= 123.18 + 2(-173.33)(0.3) \\ &= \underline{19.182}\end{aligned}$$



- d. Find the turning point in the relationship between the computer-student ratio and test scores.

$$\text{Turning pt} \Rightarrow \frac{\Delta \text{Score}}{\Delta \text{CSR}} = 0$$

$$\therefore \hat{\beta}_1 + 2\hat{\beta}_2 \cdot \text{CSR} = 0$$

$$\therefore \text{CSR}^* = -\frac{\hat{\beta}_1}{2\hat{\beta}_2} = \frac{-123.18}{2 \cdot (-173.33)} = 0.36$$

- e. Refer to the following information and your answer in b. Do you think the turning point is relevant? Explain.

. su comp_stu

Variable	Obs	Mean	Std. dev.	Min	Max
comp_stu	420	.1359266	.0649558	0	.4208333

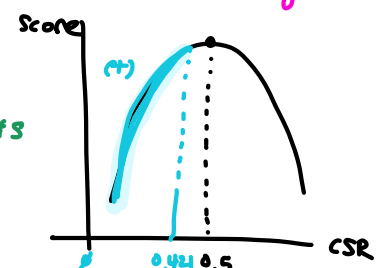
Yes. For school districts w/ a CSR below the turning pt (0.36),

adding on add'l comp per student increases test scores, on avg.

However, for districts w/ a CSR above the turning pt (btw 0.36 & 0.421), adding one more comp per student decreases test scores, on avg.

= What if the turning pt were 0.5?

⇒ No. The turning pt is NOT relevant b/c there are no districts w/ a CSR above 0.5. Therefore, in practice, an add'l comp per student only has a positive effect on test scores.



- f. Conduct a significance test at the 5% level to evaluate whether the relationship between the computer-student ratio and test scores is linear. You may use the information in the Stata output.

$$\text{Score}_i = \beta_0 + \beta_1 \text{CSR}_i + \beta_2 \text{CSR}_i^2 + \beta_3 \text{STR}_i + u_i$$

$$\begin{aligned} \textcircled{1} \quad & H_0: \beta_2 = 0 \equiv \beta_{2,0} \leftarrow (\text{linear}) \\ & H_a: \beta_2 \neq 0 \leftarrow (\text{non linear}) \end{aligned}$$

$$\textcircled{2} \quad \text{sig. lv } \alpha = 0.05$$

$$\textcircled{3} \quad \text{test statistic} = \frac{\hat{\beta}_2 - \beta_{2,0}}{\text{se}(\hat{\beta}_2)} = \frac{\hat{\beta}_2}{\text{se}(\hat{\beta}_2)} = \boxed{-1.32}$$

Stata table

$$\begin{aligned} \textcircled{4} \quad \text{p-value} &= P(|t| > |\text{test statistic}|) \\ &= P(|t| > 1.32) \\ &= \boxed{0.188} \end{aligned}$$

⑤ conclusion

$$\text{p-value} \quad \underline{vs} \quad \alpha$$

$$0.188 > 0.05 \Rightarrow \text{We fail to reject } H_0$$

that CSR and scores have "linear" relationship.

Appendix: Let's practice interpreting Stata's regression results

We run the following regression model using school district-level data:

$$\text{Testscore}_i = \beta_0 + \beta_1 \text{CompStudent}_i + \beta_2 \text{CompStudent}_i^2 + \beta_3 \text{STR}_i + u_i, \text{ where } i \text{ represents the school district,}$$

testscr: Average test scores

str: Student-Teacher Ratio

comp_stu: Computer-Student Ratio

comp_stu2: Computer-Student Ratio squared

testscr	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
comp_stu	123.1777	46.28552	2.66	0.008	32.19501	214.1603
comp_stu2	-173.3259	131.4739	-1.32	0.188	-431.7619	85.11018
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Q1. Fill out the table

	Coefficient	Estimated Coefficient	Standard error of estimated coefficient	Test statistic (t-statistic) of the estimated coefficient	p-value of the estimated coefficient	A 95% CI for the parameter [LB, UB]
CompStudent	β_1	$\widehat{\beta}_1 =$	$se(\widehat{\beta}_1) =$			[,]
STR	β_3					
Constant (y-intercept)						

Note. CI= Confidence Interval; LB=Lower Bound; UB=Upper Bound

Q2. Calculate the test statistic for $\widehat{\beta}_3$ manually and compare it with the test statistic shown in the Stata table.

Q3. Is the estimated coefficient on *STR* statistically significant at the 5% level?

ANSWERS

Q1.

	Coefficient (Parameter)	Estimated Coefficient	SE of estimated coeff	Test statistic (t-statistic) of the estimated coefficient	p-value of the estimated coefficient	A 95% CI for the parameter [LB, UB]
CompStudent	β_1	$\widehat{\beta}_1 = 123.1777$	$se(\widehat{\beta}_1) = 46.28552$	2.66	0.008	[32.19501 , 214.1603]
STR	β_3	$\widehat{\beta}_3 = -1.533944$	$se(\widehat{\beta}_3) = .4943405$	-3.10	0.002	[-2.505661 , -.5622273]
Constant (y-intercept)	β_0	$\widehat{\beta}_0 = 671.4727$	$se(\widehat{\beta}_0) = 11.16598$	60.14	0.000	[649.5239 , 693.4214]

Q2. Test-statistic = $\frac{\widehat{\beta}_3}{se(\widehat{\beta}_3)} = \frac{-1.533944}{.4943405} = -3.10$

Q3. Yes. The p-value of $\widehat{\beta}_3$ is 0.002, which is smaller than 0.05. Therefore, the estimated coefficient is statistically significant at the 5% level. In other words, we reject the null hypothesis that $\beta_3 = 0$, implying that there is a relationship between the ~~computer student ratio~~ **STR** and test scores.